

Computer Vision Final Project Report

Foundational Feature Extraction

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1 Activity A: Foundational Algorithm (Harris Corner Detector)

To demonstrate a rigorous understanding of continuous mathematics translated into discrete array operations, the Harris Corner Detector was implemented from scratch utilizing standard Python data structures and NumPy arrays.

1.1 Mathematical Formulation

The algorithm identifies distinctive points by analyzing local intensity changes. First, the image gradients I_x and I_y are computed using custom 2D convolutions with Sobel operators. The structure tensor M is then constructed for a local neighborhood:

$$M = \sum_{x,y} w(x,y) \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix} \quad (1)$$

where $w(x,y)$ represents a Gaussian smoothing window.

The Harris response function R is computed to evaluate corner strength without explicitly computing eigenvalues:

$$R = \det(M) - k \cdot \text{trace}(M)^2 \quad (2)$$

where $\det(M) = \lambda_1 \lambda_2$ and $\text{trace}(M) = \lambda_1 + \lambda_2$, with k being an empirical constant set to 0.04.

1.2 Implementation Details

To strictly adhere to the project requirements, high-level filtering functions were avoided. A custom convolution engine was written to apply the Sobel and Gaussian kernels. Furthermore, the Non-Maximal Suppression (NMS) was implemented manually by iterating through the response map and extracting local maxima within a defined neighborhood, isolating the exact pixel coordinates of each corner.

1.3 Results

Below is the output of the algorithm applied to the test image.

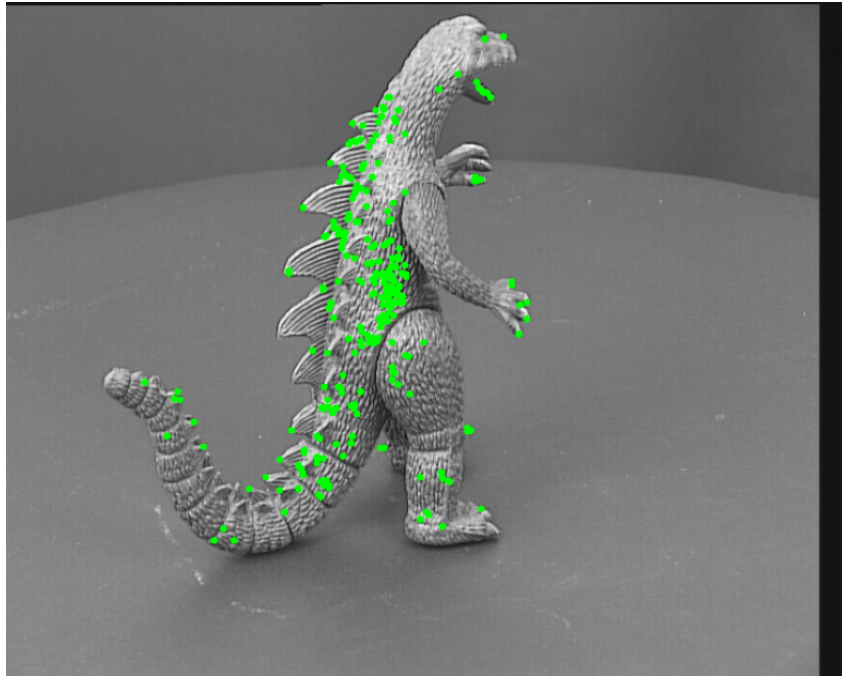


Figure 1: Corners detected using the custom Harris implementation.